

- Instant messaging, with contact lists
- Peer-to-peer file sharing of MP3s, pioneered by Napster

Interestingly, the first two killer applications came from the research community, whereas the last two were created by a few young entrepreneurs.

The period from 1995 to 2001 was a roller-coaster ride for the Internet in the financial markets. Before they were even profitable, hundreds of Internet startups made initial public offerings and started to be traded in a stock market. Many companies were valued in the billions of dollars without having any significant revenue streams. The Internet stocks collapsed in 2000–2001, and many startups shut down. Nevertheless, a number of companies emerged as big winners in the Internet space, including Microsoft, Cisco, Yahoo, e-Bay, Google, and Amazon.

### 1.7.5 The New Millennium

Innovation in computer networking continues at a rapid pace. Advances are being made on all fronts, including deployments of faster routers and higher transmission speeds in both access networks and in network backbones. But the following developments merit special attention:

- Since the beginning of the millennium, we have been seeing aggressive deployment of broadband Internet access to homes—not only cable modems and DSL but also fiber to the home, as discussed in Section 1.2. This high-speed Internet access has set the stage for a wealth of video applications, including the distribution of user-generated video (for example, YouTube), on-demand streaming of movies and television shows (e.g., Netflix), and multi-person video conference (e.g., Skype).
- The increasing ubiquity of high-speed (54 Mbps and higher) public WiFi networks and medium-speed (up to a few Mbps) Internet access via 3G and 4G cellular telephony networks is not only making it possible to remain constantly connected while on the move, but also enabling new location-specific applications. The number of wireless devices connecting to the Internet surpassed the number of wired devices in 2011. This high-speed wireless access has set the stage for the rapid emergence of hand-held computers (iPhones, Androids, iPads, and so on), which enjoy constant and untethered access to the Internet.
- Online social networks, such as Facebook and Twitter, have created massive people networks on top of the Internet. Many Internet users today “live” primarily within Facebook. Through their APIs, the online social networks create platforms for new networked applications and distributed games.

- As discussed in Section 1.3.3, online service providers, such as Google and Microsoft, have deployed their own extensive private networks, which not only connect together their globally distributed data centers, but are used to bypass the Internet as much as possible by peering directly with lower-tier ISPs. As a result, Google provides search results and email access almost instantaneously, as if their data centers were running within one's own computer.
- Many Internet commerce companies are now running their applications in the "cloud"—such as in Amazon's EC2, in Google's Application Engine, or in Microsoft's Azure. Many companies and universities have also migrated their Internet applications (e.g., email and Web hosting) to the cloud. Cloud companies not only provide applications scalable computing and storage environments, but also provide the applications implicit access to their high-performance private networks.

## 1.8 Summary

In this chapter we've covered a tremendous amount of material! We've looked at the various pieces of hardware and software that make up the Internet in particular and computer networks in general. We started at the edge of the network, looking at end systems and applications, and at the transport service provided to the applications running on the end systems. We also looked at the link-layer technologies and physical media typically found in the access network. We then dove deeper inside the network, into the network core, identifying packet switching and circuit switching as the two basic approaches for transporting data through a telecommunication network, and we examined the strengths and weaknesses of each approach. We also examined the structure of the global Internet, learning that the Internet is a network of networks. We saw that the Internet's hierarchical structure, consisting of higher- and lower-tier ISPs, has allowed it to scale to include thousands of networks.

In the second part of this introductory chapter, we examined several topics central to the field of computer networking. We first examined the causes of delay, throughput and packet loss in a packet-switched network. We developed simple quantitative models for transmission, propagation, and queuing delays as well as for throughput; we'll make extensive use of these delay models in the homework problems throughout this book. Next we examined protocol layering and service models, key architectural principles in networking that we will also refer back to throughout this book. We also surveyed some of the more prevalent security attacks in the Internet day. We finished our introduction to networking with a brief history of computer networking. The first chapter in itself constitutes a mini-course in computer networking.

So, we have indeed covered a tremendous amount of ground in this first chapter! If you're a bit overwhelmed, don't worry. In the following chapters we'll revisit all of these ideas, covering them in much more detail (that's a promise, not a threat!). At this point, we hope you leave this chapter with a still-developing intuition for the pieces